

Introduction to Structural Equation Modeling using lavaan

Intro and regression

R. M. Kuiper (and many others)

Department of Methodology & Statistics
Utrecht University

Outline of this lecture

SEM

SAPI

Lavaan commands

Regression

Path model

Steps

Technical output

Steps Ctd.

Model fit

The end

Table of Contents

SEM

SAPI

Lavaan commands

Regression

Path model

Steps

Technical output

Steps Ctd.

Model fit

The end

Path Diagram: Graphical representation of SEM

y1

Observed variable

F1

Construct = latent (unmeasured) variable / factor



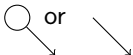
Direct relationship



Covariance

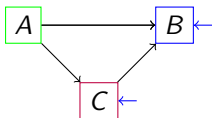


Variance

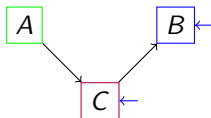


Residual variance / measurement error

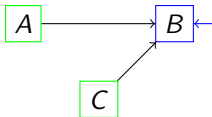
Path models



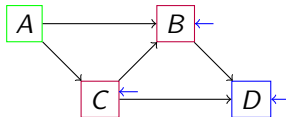
Partial mediation



Full mediation

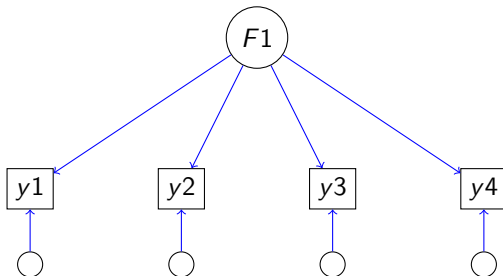


Multiple regression



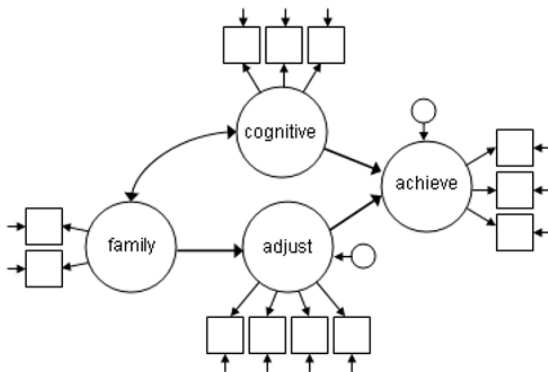
More complex path models

Measurement models



SEM

- From theory to model (and path diagram)
- Compare



Example: South African Personality Inventory Project (SAPI)



Carin Hill
Leon Jackson
Deon Meiring
J. Aleweyn Nel

Ian Rothmann
Michael Temane
Velichko H. Valchev
Fons J. R. van de Vijver

- Selection of 1000 participants.
- Selection of 7 out of 262 personality items:
 - Regression:
 - Outcome: Q77, I enjoy telling funny stories
 - Predictor: Age
 - Path model:
 - Outcome: Q196, I make others laugh
 - Mediator: Q77, I enjoy telling funny stories
 - Predictor: Age
- Later, we use: Age, ReadAb, Q44, Q63, and Q76.

Table of Contents

SEM

SAPI

Lavaan commands

Regression

Path model

Steps

Technical output

Steps Ctd.

Model fit

The end

1. Loading data into R

For example, in case of a .txt file:

```
data_sapi <- read.table("Sapi.txt", header = T)

data_sapi[apply(data_sapi,
  function(x) as.character(x) %in% c("-999") )] <- NA
```

3. Draw your model



4. Specify your model



- Translate $\rightarrow$ into $\sim$.
- Translate 'residual variance notation' into $\sim\sim$.
Note = residual variance = a variance = special case covariance.
- Translate $\leftrightarrow$ (= covariance) into $\sim\sim$.
- Do not forget about intercepts: Translate into $\sim$.
- Translate 'circled arrow' (= variance) into $\sim\sim$.
Note = variance = special case covariance.
- Do not forget about means: Translate into $\sim$.

4. Specify your model: regression SAPI



- Translate $\rightarrow$ into $\sim$.
- Translate 'residual variance notation' into $\sim\sim$.
Note = residual variance = a variance = special case covariance.
- Translate $\leftrightarrow$ (= covariance) into $\sim\sim$.
- Do not forget about intercepts: Translate into $\sim$.

```
mod.regr <- '
  Q77 ~ Age # regression
  Q77 ~~ Q77 # residual variance
  Q77 ~ 1    # intercept
'
```

4. Specify your model: regression general

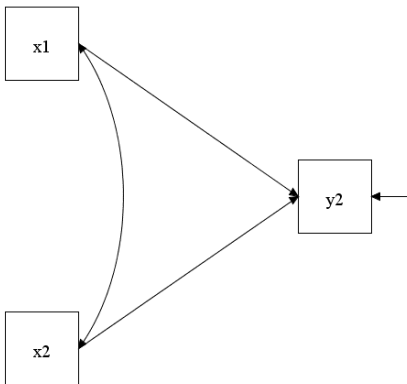
A regression model is specified as follows:
 $\text{dependent} \sim \text{predictor1} + \text{predictor2} + \text{etc.}$

Additionally, one needs to specify that the error (of the dependent variable) has a variance:
 $\text{dependent} \sim\sim \text{dependent}$

If needed, one can specify to include an intercept:
 $\text{dependent} \sim 1$

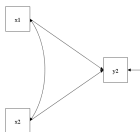
Note that one include intercept directly:
 $\text{dependent} \sim 1 + \text{predictor1} + \text{predictor2} + \text{etc.}$

4. Specify your model: regression DIY



- Do you recognize this model?
- Specify this in lavaan code.
Do not forget the parameters that are not drawn, but do are there.

4. Specify your model: regression answer 2



```
mod.regr_DIY <- '
  y2 ~ 1 + x1 + x2  # intercept + regression
  y2 ~~ y2          # residual variance

# Make it a habit to also specify:
x1 + x2 ~ 1         # means predictors - DO specify this!
x1 + x2 ~~ x1       # (co)variance predictors
x2 ~~ x2            # variance x2
'
```

Note: As opposed to what I say in the video, do specify the means for the predictors explicitly; otherwise they will be set to 0.

4. Specify your model: regression answer general

```
y ~ 1 + x_1 + x_2 + ... + x_k # intercept + regression
y ~~ y                        # residual variance
```

```
# (co)variances of exogenous variables:
```

```
x_1 + x_2 + ... + x_k ~~ x_1
      x_2 + ... + x_k ~~ x_2
                        ...
x_k   ~~ x_k
```

5. Fit the regression model

```
fit.regr <- lavaan(model = mod.regr, data = data_sapi)
```

5. Fit the regression model: All code

```
# Data
data_sapi <- read.table("Sapi.txt", header = T)
data_sapi[sapply(data_sapi,
  function(x) as.character(x) %in% c("-999") )] <- NA

# Model
mod.regr <- '
  Q77 ~ Age # regression
  Q77 ~~ Q77 # residual variance
  Q77 ~ 1   # intercept
'

# Fit model
fit.regr <- lavaan(model = mod.regr, data = data_sapi)
```

6. Plot the lavaan model in R

- lavaanPlot
- graph_sem from tidySEM
- semPaths from semPlot
- probably many more...

Plot with lavaanPlot

```
if (!require("lavaanPlot")) install.packages("lavaanPlot")
library(lavaanPlot)
```

```
lavaanPlot(model = fit.regr,
  graph_options = list(overlap = "true",
    fontsize = "20"),
  node_options = list(shape = "box",
    fontname = "Helvetica"),
  edge_options = list(color = "grey"),
  coefs = T, stand = T, covs = T,
)
```

Plot with tidySEM

```
library(tidySEM)  
if (!require("tidySEM")) install.packages("tidySEM")
```

```
graph_sem(fit.regr)
```

Plot with semPlot

```
if (!require("semPlot")) install.packages("semPlot")
library(semPlot)
```

```
semPaths(fit.regr, "par", weighted = FALSE, nCharNodes = 7,
         shapeMan = "rectangle", sizeMan = 8, sizeMan2 = 5)
```

9. Acquiring the summary

```
summary(fit.regr, fit.measures = TRUE,
        ci = TRUE, rsquare = TRUE)
```

```
parameterEstimates(fit.regr)
```

```
inspect(fit.regr, 'r2')
# As an example, there are more.
```

```
fitMeasures(fit.regr,
             c("npar", "logl", "unrestricted.logl",
               "chisq", "pvalue",
               "cfi", "tli", "rmsea"))
# As an example, there are more.
```

Table of Contents

SEM

SAPI

Lavaan commands

Regression

Path model

Steps

Technical output

Steps Ctd.

Model fit

The end

Analyzing the SAPI data: Regression

Variables:

- Outcome: Q77, I enjoy telling funny stories
- Predictor: Age



Specifying and fitting regression model in lavaan

```
mod.regr <- '
  Q77 ~ Age # regression
  Q77 ~~ Q77 # residual variance
  Q77 ~ 1   # intercept
'
```

```
fit.regr <- lavaan(model = mod.regr, data = data_sapi)
```

Plot with lavaanPlot

```
lavaanPlot(model = fit.regr,  
            graph_options = list(overlap = "true",  
                                  fontsize = "20"),  
            node_options = list(shape = "box",  
                                  fontname = "Helvetica"),  
            edge_options = list(color = "grey"),  
            coefs = T, stand = F, covs = T,  
            )
```

```
## Error in 'loadNamespace()':  
## ! there is no package called 'webshot'
```

```
# Note: I use stand = F here.
```

Summary regression model in lavaan

```
summary(fit.regr, fit.measures = TRUE,  
        ci = TRUE, rsquare = TRUE)
```

Estimates:

```
parameterEstimates(fit.regr)[c(1,2,3),] # or ...[1:3,]
```

##	lhs	op	rhs	est	se	z	pvalue	ci.lower	ci.upper
## 1	Q77	~	Age	-0.022	0.004	-5.275	0	-0.030	-0.014
## 2	Q77	~~	Q77	1.184	0.068	17.536	0	1.052	1.317
## 3	Q77	~1		4.275	0.135	31.657	0	4.011	4.540

Regression equation

```
parameterEstimates(fit.regr)[c(1,2,3),] # or ...[1:3,]
```

##	lhs	op	rhs	est	se	z	pvalue	ci.lower	ci.upper
## 1	Q77	~	Age	-0.022	0.004	-5.275	0	-0.030	-0.014
## 2	Q77	~~	Q77	1.184	0.068	17.536	0	1.052	1.317
## 3	Q77	~1		4.275	0.135	31.657	0	4.011	4.540

General regression equation (for person i):

$Q77_i = B_0 + B_1 * Age_i + e_i$, with $e_i \sim N(0, \sigma^2)$.

Or: $\hat{Q77}_i = B_0 + B_1 * Age_i$.

Regression equation (for person i):

$Q77_i = 4.275 - 0.022 * Age_i + e_i$, with $e \sim N(0, 1.184)$.


```
inspect(fit.regr, 'r2')
```

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

Intermezzo: Correlation Age and Q77 and R^2 in R

```
corr <- cor(data_sapi[, 2], data_sapi[, 9],
             use = "complete.obs")

round(corr, 3)    # = correlation between Age and Q77

## [1] -0.208
```

```
round(corr^2, 3) # =  $R^2$ 

## [1] 0.043

#  $R^2$  = proportion of shared/explained variance.
# This is the  $R^2$  in a regression with one predictor.
```

In a regression with one outcome and one predictor,
the R^2 is the square of their correlation.

Intermezzo: Correlation Age and Q77 and R^2 in R Ctd.

Remark:

I would not use lavaan for correlations, but you can:

```
mod.corr <- '  
  Q77 ~~ Q77  
  Age ~~ Age  
  Q77 ~~ Age # covariance  
  # If standardized, then correlation  
'  
  
fit.corr <- lavaan(model = mod.corr, data = data_sapi)  
parameterEstimates(fit.corr, stand=T)[3,'std.all']  
  
## [1] -0.2080665
```

Message intermezzo:

In a regression with one outcome and one predictor,
the R^2 is the square of their correlation.

Remark assumptions

We do not cover checking assumptions in the course,
but it is important to do.

You can find more information online; e.g.:

- <https://stats.stackexchange.com/questions/164261/sem-structural-equation-modelling-assumptions>
- https://www.regorz-statistik.de/blog/lavaan_linearity.html

Table of Contents

SEM

SAPI

Lavaan commands

Regression

Path model

Steps

Technical output

Steps Ctd.

Model fit

The end

Analyzing the SAPI data: Path model (Mediation)

Variables:

- Outcome: Q196, I make others laugh
- Mediator: Q77, I enjoy telling funny stories
- Predictor: Age



Specification path model in lavaan

```

model.path <- '
  # regressions
  Q77 ~ Age
  Q196 ~ Q77

  # residual variances
  Q77 ~~ Q77
  Q196 ~~ Q196

  # variance of predictor
  Age ~~ Age
  # mean of predictor
  Age ~ 1

  ## intercepts
  Q77 ~ 1
  Q196 ~ 1
'

```

Plot with lavaanPlot

```
lavaanPlot(model = fit_path,  
            graph_options = list(overlap = "true",  
                                  fontsize = "20"),  
            node_options = list(shape = "box",  
                                  fontname = "Helvetica"),  
            edge_options = list(color = "grey"),  
            coefs = T, stand = F, covs = T,  
            ) # Note: stand = F
```

```
## Error in 'loadNamespace()':  
## ! there is no package called 'webshot'
```

Summary path model in lavaan

```
summary(fit_path, fit.measures = TRUE,
        ci = TRUE, rsquare = TRUE)
```

Estimates

```
parameterEstimates(fit_path)[,-7] # i.e., without p-value
```

##	lhs	op	rhs	est	se	z	ci.lower	ci.upper
## 1	Q77	~	Age	-0.021	0.004	-4.971	-0.029	-0.013
## 2	Q196	~	Q77	0.451	0.027	16.724	0.398	0.504
## 3	Q77	~~	Q77	1.179	0.067	17.479	1.047	1.312
## 4	Q196	~~	Q196	0.545	0.031	17.479	0.484	0.606
## 5	Age	~~	Age	108.692	6.219	17.479	96.504	120.881
## 6	Age	~1		30.612	0.422	72.580	29.785	31.439
## 7	Q77	~1		4.248	0.136	31.175	3.981	4.516
## 8	Q196	~1		2.153	0.102	21.167	1.954	2.352

```
##      Q77      Q196
## 0.039 0.314
```

Table of Contents

SEM

SAPI

Lavaan commands

Regression

Path model

Steps

Technical output

Steps Ctd.

Model fit

The end

Step by step (repitition): Step 1

Load data into R

Step 2

Data screening

e.g., check measurement levels, missing data (notation), and correlations.

Not part of lavaan,
plenty of information on the internet.

Step 3

Draw your model

Step 4

Specify your model into lavaan syntax
i.e., Translate your drawing into lavaan syntax

Step 5

Fit the model with lavaan

Step 6

Plot the lavaan model

Step 7

Ensure model specification is correct
using

- Plots (for example, using lavaanPlot)
- Technical output (using lavInspect),
discussed in next intermezzo.

Step 8

Check model assumptions

Not part of lavaan,
plenty of information on the internet.

Step 9

Acquire the summary from lavaan

Intermezzo: Technical output





Degrees of freedom

Degrees of freedom (df) =

Number of unique elements - Number of model parameters =
knowns - unknowns.

$df = ((p(p + 1)/2) + p) - npar$,
with p the number of variables in the model.

Number of model parameters - SAPI regression

Number of model parameters (unknowns):

```
fitMeasures(fit.regr, "npar")

## npar
##      3
```

Which ones can be inspected in the technical output:

```
lavInspect(fit.regr)
```

More explanation later on.

For now: 3 model parameters estimated.

Number of unique elements

Note: Raw data can be summarized by

- a variance matrix (S)
- a mean vector (m)

Number of unique elements (knowns):

- Unique elements in variance matrix = $p(p + 1)/2$
- Possibly: Unique elements in mean = p

with p the number of variables in the model.

In SAPI regression ($p = 2$):

- Unique elements S : $p(p + 1)/2 = 2 * 3/2 = 3$
- Unique elements m : $p = 2$

Number of model parameters = $3 + 2 = 5$.

BUT:

0

Lavaan defaults

```
parameterEstimates(fit.regr)[,1:7] # without ci
```

##	lhs	op	rhs	est	se	z	pvalue
## 1	Q77	~	Age	-0.022	0.004	-5.275	0
## 2	Q77	~~	Q77	1.184	0.068	17.536	0
## 3	Q77	~1		4.275	0.135	31.657	0
## 4	Age	~~	Age	111.304	0.000	NA	NA
## 5	Age	~1		30.706	0.000	NA	NA

By default, lavaan secretly estimates

- the means and variances of exogenous observed variables
- the covariances between exogenous variables

as does Mplus.

SAPI regression: Fully specified

```
mod.regr_full <- '  
  Q77 ~ Age # regression  
  Q77 ~~ Q77 # residual variance  
  Q77 ~ 1   # intercept  
  
  # (co)variances and means predictor(s)  
  Age ~~ Age # variance predictor  
  Age ~ 1    # mean predictor  
  #  
  # Estimated by lavaan by default,  
  # but now also part of lavInspect()  
  
'  
fit.regr_full <- lavaan(model = mod.regr_full,  
                        data = data_sapi)
```

SAPI regression: Same estimates of course

```
parameterEstimates(fit.regr)[,1:7]
```

##	lhs	op	rhs	est	se	z	pvalue
## 1	Q77	~	Age	-0.022	0.004	-5.275	0
## 2	Q77	~~	Q77	1.184	0.068	17.536	0
## 3	Q77	~1		4.275	0.135	31.657	0
## 4	Age	~~	Age	111.304	0.000	NA	NA
## 5	Age	~1		30.706	0.000	NA	NA

```
parameterEstimates(fit.regr_full)[,1:7]
```

##	lhs	op	rhs	est	se	z	pvalue
## 1	Q77	~	Age	-0.022	0.004	-5.275	0
## 2	Q77	~~	Q77	1.184	0.068	17.536	0
## 3	Q77	~1		4.275	0.135	31.657	0
## 4	Age	~~	Age	111.304	6.347	17.536	0
## 5	Age	~1		30.706	0.425	72.177	0

Technical output

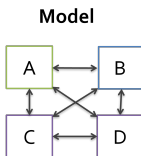
Which model parameters are estimated can be inspected in the technical output:

```
lavInspect(fit.regr_full)
```

```
## $lambda
##      Q77 Age
## Q77    0   0
## Age    0   0
##
## $theta
##      Q77 Age
## Q77    0
## Age    0   0
##
## $psi
##      Q77 Age
## Q77    2
## Age    0   4
```

```
## $beta
##      Q77  Age
## Q77    0    1
## Age    0    0
##
## $nu
##      intrcp
## Q77        0
## Age        0
##
## $alpha
##      intrcp
## Q77        3
## Age        5
```


- ◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

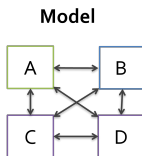


- Unique elements S : $p(p+1)/2 = 4 * 5/2 = 10$
- Unique elements m : $p = 4$

$$10+4 = 14$$

Saturated model ($p = 4$) - with means

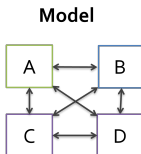
Data					
		A	B	C	D
S =	A	σ^2_A			
	B	σ_{BA}	σ^2_B		
	C	σ_{CA}	σ_{BC}	σ^2_C	
	D	σ_{DA}	σ_{DB}	σ_{DC}	σ^2_D



Number of model parameters:

- cross relations / covariances: 6
- variances: 4
- means: 4

$$(6+4)+4 = 14$$



Hence, $df = 14 - 14 = 0$.

Example saturated model ($p = 4$) Ctd.

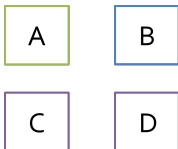
```
fitMeasures(fit_sat,  
            c("npar", "logl", "unrestricted.logl",  
              "chisq", "pvalue",  
              "cfi", "tli", "rmsea"))
```

##	npar	logl	unrestricted.logl
##	14.000	-4498.184	-4498.184
##	pvalue	cfi	tli
##	NA	1.000	1.000

Note: At the end, more details about model fit.

Baseline model ($p = 4$)

- Variables: A, B, C, and D
 - How many unique elements?
 - $4(4+1)/2 + 4 = 14$
- Baseline model = Unrelated variables
 - How many model parameters?
 - 4 variances and 4 means = 8 parameters
- How many df?
 - $df = 14 - 8 = 6$



Example baseline model ($p = 4$) Ctd.

```
fitMeasures(fit_base4,  
  c("npar", "logl", "unrestricted.logl",  
    "chisq", "pvalue",  
    "cfi", "tli", "rmsea"))  
  
##              npar              logl  unrestricted.logl  
##              8.000             -4530.256             -4498.184  
##              pvalue              cfi              tli  
##              0.000              0.000              0.000
```

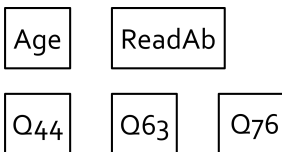
No deviation from the baseline model,
because we estimated it: CFI = 0 and TLI = 0.

Note: At the end, more details about model fit.

- ◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ▶ ↺ 🔍 ↻

Baseline model ($p = 5$): DIY

- Variables: A, B, C, D, and E
 - How many unique elements?
 -
- Baseline model = Unrelated variables
 - How many model parameters?
 -
- How many df?
 -



Baseline model ($p = 5$): Answer

- Variables: A, B, C, D, and E
 - How many unique elements?
 - $5(5+1)/2 + 5 = 20$
- Baseline model = Unrelated variables
 - How many model parameters?
 - 5 variances and 5 means = 10 parameters
- How many df?
 - $df = 20 - 10 = 10$

Age

ReadAb

Q44

Q63

Q76

```
lavInspect(fit_base5)$theta
```

```
lavInspect(fit_base5)$nu
```

Example baseline model ($p = 5$) Ctd.

```
fitMeasures(fit_base5,
  c("npars", "logl", "unrestricted.logl",
    "chisq", "pvalue",
    "cfi", "tli", "rmsea"))
```

	npars	logl	unrestricted.logl
	10.000	-5440.660	-5360.694
	pvalue	cfi	tli
	0.000	0.000	0.000

Note: At the end, more details about model fit.

Table of Contents

SEM

SAPI

Lavaan commands

Regression

Path model

Steps

Technical output

Steps Ctd.

Model fit

The end

Step by step - there are more: Step 10

Understand all errors/warnings

Note: Not all warning messages have to be fixed,
but know what the problem is.

What to do with missing data?

11. Handling missing data: FIML

Note: Will be discussed in detail in Missing Data lecture.

Model needs to be fully specified (elsewise error):

```
mod.regr_full <- '
  Q77 ~ Age # regression
  Q77 ~~ Q77 # residual variance
  Q77 ~ 1   # intercept

  # (co)variances and means predictor(s)
  Age ~~ Age # variance predictor
  Age ~ 1    # mean predictor
'

fit.regr_fiml <- lavaan(model = mod.regr_full, data = data_sapi,
                        missing='fiml', fixed.x=FALSE)

## Warning: lavaan->lav_data_full():
##      some cases are empty and will be ignored:  453 535.
```

11. Handling missing data: FIML Ctd.

Previous estimates

```
parameterEstimates(fit.regr)[c(1,2,3),]
```

##	lhs	op	rhs	est	se	z	pvalue	ci.lower	ci.upper
## 1	Q77	~	Age	-0.022	0.004	-5.275	0	-0.030	-0.014
## 2	Q77	~~	Q77	1.184	0.068	17.536	0	1.052	1.317
## 3	Q77	~1		4.275	0.135	31.657	0	4.011	4.540

FIML estimates

```
parameterEstimates(fit.regr_fiml)[c(1,2,3),]
```

##	lhs	op	rhs	est	se	z	pvalue	ci.lower	ci.upper
## 1	Q77	~	Age	-0.021	0.004	-5.341	0	-0.029	-0.013
## 2	Q77	~~	Q77	1.133	0.052	21.829	0	1.031	1.234
## 3	Q77	~1		4.219	0.126	33.524	0	3.972	4.465

Step 12

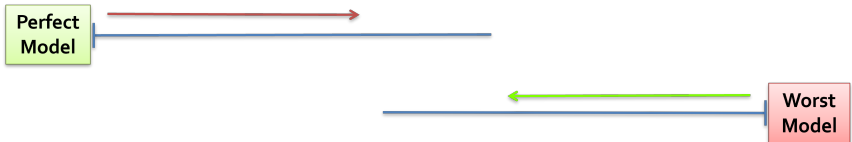
Are you really really sure that everything went well?

Model Test and Fit Indices

- Log-likelihood ratio (Chi-Square) test
- RMSEA
- CFI/TLI
- Chi-Square Test of Model Fit for the Baseline Model
- SRMR
- Information Criteria: AIC, BIC

A good fit index...

- Not sensitive to sample size
- Sensitive to discrepancies between modeled data and observed data



- Reflects the degree of parsimony of the model

Log-likelihood ratio test

- $-2\log\frac{L_0}{L_a} = -2\log L_0 + 2\log L_a$
- Statistic is Chi-square distributed
- Model test: unrestricted model is model that led to sample variance-covariance matrix
- $H_0 : \chi^2 = 0$ (perfect fit)



- + actual test for fit
- sensitive to sample size
- inflated with non-normal data

RMSEA (Root Mean Square Error of Approximation)

$$RMSEA = \sqrt{\frac{\chi^2 - df}{(n-1)df}}$$

= bias-corrected estimate of divergence of model-implied and population var-covar matrix.



Absolute index:

- Index of discrepancy between model and data
 - + sensitive to model complexity, namely: bias correction by ' $-df$ ' (in ' $\chi^2 - df$ ', in formula above))
 - - less suitable with small sample size (with low df)
- Rule of thumb: $<.08$ (mediocre), $<.05$ (close / good)
- Inspect CIs

Relative indices: AIC/BIC

- Information theoretic:
Combine goodness of fit χ^2 and parsimony (nr. of parameters k).



- AIC (Akaike Information Criterion)
 - $AIC = \chi^2 + 2k$
- BIC (Bayesian Information Criterion)
 - $BIC = \chi^2 + k * \ln(n)$
- No rule of thumb, values depend on actual data set.
- Choose the model with the **lowest** AIC and/or BIC.

Which to use?

- Report several indices of different types (CFI, RMSEA)
- All others than chi-square should look good
- Improve the model (exploratory!) by
 - removing n.s. paths and/or
 - adding significant paths,
 - see <https://lavaan.ugent.be/tutorial/modindices.html>,but **only if**: it makes theoretical sense, **and**: you report your modifications



Model Comparison

- Nested models (i.e., one model is a special case of the other):
Chi-square difference test / AIC / BIC
- Non-nested models:
AIC / BIC

In lavaan: use `anova(fit.model1, fit.model2)`.

If and only if the covariance matrix is equal over models

=

same participants, same variables!

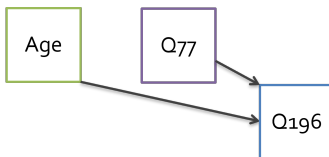
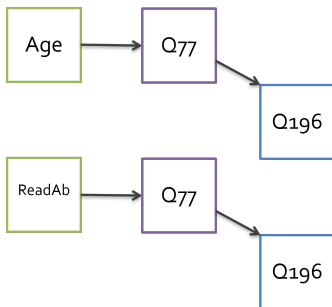
Model Comparisons: Nested models

Nested models: one model is a special case of the other.



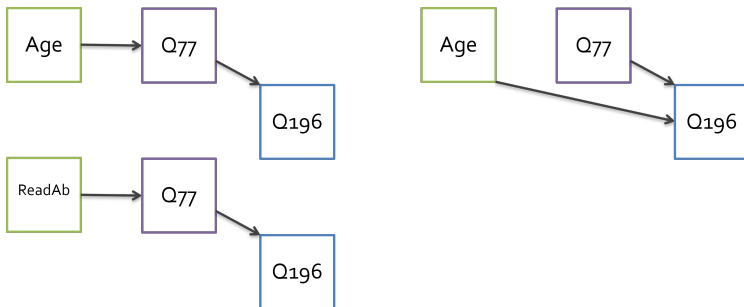
Model Comparisons: Nested models?

- Nested?
- Chi-square test vs AIC/BIC?



Model Comparisons: Nested models? - Answers

- Nested? No
- Chi-square test vs AIC/BIC? AIC/BIC



Model Comparisons: AIC / BIC

Suitable for comparing

- both nested and non-nested models.
- more than two models.

Choose the model with the **lowest** AIC and/or BIC.

To be discussed in the Model Selection lecture

- Quantify relative support via IC weights.

```
library(devtools) # Make sure you have Rtools
install_github("rebeccakuiper/ICweights")
library(ICweights) # ?IC.weights
# Or:
library(restriktor) # ?calc_ICweights
```

- Extension AIC for order-restricted hypotheses (GORIC and GORICA).

```
library(restriktor) # ?goric
```

Table of Contents

The end

Summary

- Short SEM overview
- Simple analyses in lavaan
- Step-by-step checklist
- Model fit

Take home message

- Know what's happening:
 - Check whether unique elements - estimated = df
 - Get to know lavaan defaults
 - Read warnings (!)
- Make use of FIML (if possible)

Thanks & How to proceed

Thanks for listening!

Are there any questions?

- Ask fellow participant on course platform.
- Ask teacher during Q&A (or via course platform).
- See if making the lab exercises help.
- Check the lavaan tutorial: e.g.,
<https://lavaan.ugent.be/tutorial/inspect.html>.
- Do not forget that Google is your best friend :-).

You can start working on the lab exercises.